Linear Equations & Systems

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01 • 20 FRQs + 25 MCQs • Difficulty Progression • Cross-Lesson Synthesis

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • [FRQ] • Target: Solving Multi-Step Linear Equations

Solve each linear equation for x and verify your solution by substitution:

a $5(2x-3)-4=3(x+5)$ $10x-19=3x+15 \Rightarrow 7x=34 \Rightarrow x=\frac{34}{7}$

b $\frac{3x-2}{4}+\frac{x+5}{2}=7$ $3x-2+2(x+5)=28 \Rightarrow 5x+8=28 \Rightarrow x=4$

Combine like terms on each side, isolate the variable term, and divide by the coefficient of x.

2 [Foundational] • [FRQ] • Target: Slope & Intercepts of a Line

A line passes through the points $A(-3,\,7)$ and $B(5,\,-9)$ in the xy-plane. Determine (a) the slope of the line, (b) its equation in slope-intercept form $y=mx+b$, and (c) its x-intercept.

(a) Slope $m=\frac{-9-7}{5-(-3)}=\frac{-16}{8}=-2$. (b) Using $(5,\,-9)$: $-9=-2(5)+b \Rightarrow b=1$, so $y=-2x+1$. (c) Setting $y=0$ gives $-2x+1=0 \Rightarrow x=\frac{1}{2}$, so the x-intercept is $\left(\frac{1}{2},\,0\right)$.

3 [Foundational] • [FRQ] • Target: Standard Form & Coordinate Triangle Area

Convert the standard-form linear equation $6x-8y=24$ into slope-intercept form, state the slope and both intercepts, and compute the area of the right triangle bounded by the line and the coordinate axes.

Solving for y: $-8y=-6x+24 \Rightarrow y=\frac{3}{4}x-3$. The slope is $m=\frac{3}{4}$, the y-intercept is $(0,\,-3)$, and the x-intercept is $(4,\,0)$. The right triangle with legs 4 and 3 has area $A=\frac{1}{2}(4)(3)=6$ square units.

4 [Foundational] • [FRQ] • Target: Contextual Linear Modeling

A cloud-computing server plan charges a fixed monthly maintenance fee plus a constant rate per gigabyte of outbound data transfer. Using $120\,\text{GB}$ costs 86 dollars, and using $270\,\text{GB}$ costs 131 dollars. Write a linear function $C(g)$ for the monthly cost (in dollars) of $g\,\text{GB}$, and interpret both the slope and $C(0)$ in context.

Rate per GB: $m=\frac{131-86}{270-120}=\frac{45}{150}=0.30$ dollars per GB. Using $(120,\,86)$: $86=0.30(120)+b \Rightarrow b=50$. Thus $C(g)=0.30g+50$, where 0.30 is the cost in dollars per additional gigabyte and $C(0)=50$ dollars is the fixed monthly base fee.

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5 [Foundational] • [FRQ] • Target: Parallel & Perpendicular Lines

Determine the equation of the line passing through $(-4, 6)$ that is (a) parallel to $3x + 2y = 10$, and (b) perpendicular to $3x + 2y = 10$. Write both answers in slope-intercept form.

The given line has slope $m = -\frac{3}{2}$. (a) Parallel slope $m_{\parallel} = -\frac{3}{2}$: $y - 6 = -\frac{3}{2}(x + 4) \Rightarrow y = -\frac{3}{2}x$. (b) Perpendicular slope $m_{\perp} = \frac{2}{3}$: $y - 6 = \frac{2}{3}(x + 4) \Rightarrow y = \frac{2}{3}x + \frac{26}{3}$.

6 [Foundational] • [FRQ] • Target: Isolating Quantities in Literal Equations

Rearrange the literal thermal-expansion formula $L = L_0(1 + \alpha(T - T_0))$ to express the temperature T in terms of $L, L_0, \alpha,$ and T_0.

Divide by L_0: $\frac{L}{L_0} = 1 + \alpha(T - T_0) \Rightarrow \frac{L - L_0}{L_0} = \alpha(T - T_0) \Rightarrow T = T_0 + \frac{L - L_0}{\alpha L_0}$.

7 [Foundational] • [FRQ] • Target: Conditions for No Solution vs. Infinitely Many Solutions

Find the value of the constant k for which the linear equation $4(3x - k) + 6 = 2(6x + 9)$ has infinitely many real solutions.

Expand both sides: $12x - 4k + 6 = 12x + 18$. For infinitely many solutions (an identity), the constant terms must match: $-4k + 6 = 18 \Rightarrow -4k = 12 \Rightarrow k = -3$.

Part I — Free-Response Questions (FRQs): Intermediate Modeling (8–14)

8 [Intermediate] • [FRQ] • Target: Perpendicular Line Intersection

In the xy-plane shown, line ℓ_1 has equation $y = \frac{1}{2}x + 3$ and line ℓ_2 passes through $(0, -2)$ and $(4, -10)$ so that $\ell_1 \perp \ell_2$. Find the exact coordinates of their intersection point P.

Line ℓ_2 has slope $m_2 = \frac{-10 - (-2)}{4 - 0} = -2$, so its equation is $y = -2x - 2$. Setting $\frac{1}{2}x + 3 = -2x - 2$ gives $\frac{5}{2}x = -5 \Rightarrow x = -2$, and $y = \frac{1}{2}(-2) + 3 = 2$. Thus $P = (-2, 2)$.

9 [Intermediate] • [FRQ] • Target: Linear Mixture & Concentration Modeling

An chemist mixes a 18% saline solution with a 42% saline solution to produce $600\,\text{mL}$ of a 28% saline solution. How many milliliters of each solution must be used?

Let x be the volume (in mL) of the 18% solution; then $600 - x$ is the volume of the 42% solution. Set up the mass-balance equation: $0.18x + 0.42(600 - x) = 0.28(600) \Rightarrow -0.24x + 252 = 168 \Rightarrow 0.24x = 84 \Rightarrow x = 350\,\text{mL}$ of the 18% solution and $600 - 350 = 250\,\text{mL}$ of the 42% solution.

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10 **[Intermediate]** • **[FRQ]** • **Target: Absolute Value Linear Equations**

Solve the absolute value equation $3|2x-5|-7=14$ and plot both solutions on a number line, stating the midpoint and distance from the midpoint to each solution.

Isolate the absolute value: $3|2x-5|=21 \Rightarrow |2x-5|=7$. Thus $2x-5=7 \Rightarrow x=6$, or $2x-5=-7 \Rightarrow x=-1$. The midpoint is $x=2.5$ and each solution lies at distance 3.5 from 2.5.

11 **[Intermediate]** • **[FRQ]** • **Target: Parametric Linear Equations with No Solution**

For what value(s) of the parameter a does the linear equation $(2a-5)x+3a=7x+12$ have no solution?

Rewrite as $(2a-12)x=12-3a$. A single-variable linear equation $Mx=N$ has no solution if and only if $M=0$ and $N\neq 0$. Setting $2a-12=0$ gives $a=6$, for which the right-hand side is $12-3(6)=-6\neq 0$. Thus $a=6$.

12 **[Intermediate]** • **[FRQ]** • **Target: Distance-Rate-Time Linear Modeling**

A delivery drone flies from depot A to warehouse B against a steady headwind at 36 km/h and returns along the same route with a tailwind at 54 km/h. If the total round-trip flight time is 2.5 hours, find the one-way distance d between A and B and the average speed for the entire round trip.

Total time: $\frac{d}{36}+\frac{d}{54}=2.5$. Multiply by $\operatorname{lcm}(36,54)=108$: $3d+2d=270 \Rightarrow 5d=270 \Rightarrow d=54$ km. Total distance is $2d=108$ km, so average speed is $\frac{108}{2.5}=43.2$ km/h (the harmonic mean of 36 and 54).

13 **[Intermediate]** • **[FRQ]** • **Target: Linear Function Rate of Change & Functional Equations**

A linear function f satisfies $f(2k)-f(k)=6k$ for all real k, and $f(-1)=4$. Find the formula for $f(x)$ and evaluate $f(10)$.

Since $f(x)=mx+b$, we have $f(2k)-f(k)=m(2k-k)=mk=6k$, so $m=6$. Using $f(-1)=4$: $6(-1)+b=4 \Rightarrow b=10$. Therefore $f(x)=6x+10$ and $f(10)=6(10)+10=70$.

14 **[Intermediate]** • **[FRQ]** • **Target: Piecewise Linear Tariff Modeling**

A municipal water utility bills residential customers using a two-tier piecewise linear schedule: 22 dollars base fee plus 1.80 dollars per HCF for the first 15 HCF, and 3.20 dollars per HCF for every HCF above 15. If a household's monthly bill was 81.00 dollars, how many HCF of water did the household consume?

Cost at the end of Tier 1 (15 HCF): $22+1.80(15)=22+27=49.00$ dollars. Since $81.00>49.00$, let $w>15$ be total HCF: $49.00+3.20(w-15)=81.00 \Rightarrow 3.20(w-15)=32.00 \Rightarrow w-15=10 \Rightarrow w=25$ HCF.

Part I — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 **[Challenging]** • **[FRQ]** • **Target: Perpendicular Projection & Coordinate Geometry** • **Cross-Lesson: Topic 7.01 (Triangles, Lines & Angles)**

Line ℓ has equation $3x + 4y = 24$ and intersects the x-axis at A and the y-axis at B. Find (a) the length of the altitude h from the origin $O(0, 0)$ to segment AB, and (b) the coordinates of the foot of the perpendicular H on segment AB.

(a) The intercepts are $A(8, 0)$ and $B(0, 6)$, so $AB = \sqrt{8^2 + 6^2} = 10$. By right-triangle area, $\frac{1}{2}(8)(6) = \frac{1}{2}(10)h \Rightarrow h = \frac{48}{10} = 4.8$. (b) The perpendicular line through $O(0, 0)$ has slope $\frac{4}{3}$, so $y = \frac{4}{3}x$. Substituting into $3x + 4y = 24$: $3x + \frac{16}{3}x = 24 \Rightarrow \frac{25}{3}x = 24 \Rightarrow x = \frac{72}{25} = 2.88$ and $y = \frac{96}{25} = 3.84$. Thus $H = (2.88, 3.84)$.

16 **[Challenging]** • **[FRQ]** • **Target: Linear Function Composition & Inverse** • **Cross-Lesson: Topic 2.01 (Linear Systems & Fixed Points)**

Suppose $f(x) = ax + b$ is a linear function with $a > 0$ such that the composite function satisfies $f(f(x)) = 9x - 16$ for all real x. Find the exact values of a and b, and solve $f(x) = f^{-1}(x)$.

Compute $f(f(x)) = a(ax + b) + b = a^2x + b(a + 1) = 9x - 16$. Since $a > 0$, $a^2 = 9 \Rightarrow a = 3$. Then $b(3 + 1) = -16 \Rightarrow 4b = -16 \Rightarrow b = -4$, so $f(x) = 3x - 4$. A fixed point satisfies $f(x) = x \Rightarrow 3x - 4 = x \Rightarrow x = 2$.

17 **[Challenging]** • **[FRQ]** • **Target: Parametric Intercepts & Optimization** • **Cross-Lesson: Topic 3.01 (Quadratics & Extremum Analysis)**

A line with negative slope m passes through the fixed point $P(3, 4)$ and forms a triangle in the first quadrant with the positive x- and y-axes. Express the x-intercept and y-intercept in terms of m, and verify that if $m = -\frac{4}{3}$, the area of the first-quadrant triangle is minimized at $A = 24$.

The line is $y - 4 = m(x - 3)$ with $m < 0$. Setting $y = 0$ gives the x-intercept $x_0 = 3 - \frac{4}{m}$; setting $x = 0$ gives the y-intercept $y_0 = 4 - 3m$. At $m = -\frac{4}{3}$, we obtain $x_0 = 3 - \frac{4}{-4/3} = 6$ and $y_0 = 4 - 3\left(-\frac{4}{3}\right) = 8$, giving area $A = \frac{1}{2}(6)(8) = 24$ (which is minimal since $A(m) = \frac{1}{2}\left(24 - 9m - \frac{16}{m}\right) \geq \frac{1}{2}(24 + 2\sqrt{144}) = 24$ by AM-GM).

18 **[Challenging]** • **[FRQ]** • **Target: Linear Regression Centroid & Residual Analysis** • **Cross-Lesson: Topic 6.01 (Statistics, Scatterplots & Residuals)**

In a bivariate study of $n = 8$ electric vehicles, the mean battery mass is $\bar{x} = 420\,\text{kg}$ and the mean driving range is $\bar{y} = 485\,\text{km}$. The ordinary least-squares regression line is $\hat{y} = 0.65x + b$. (a) Use the centroid property $(\bar{x}, \bar{y})$ to find b. (b) If a vehicle with a $460\,\text{kg}$ battery has an actual range of $501\,\text{km}$, compute its residual $e = y - \hat{y}$.

(a) Every OLS regression line passes through $(\bar{x}, \bar{y})$: $485 = 0.65(420) + b \Rightarrow 485 = 273 + b \Rightarrow b = 212$. (b) For $x = 460$, predicted range is $\hat{y} = 0.65(460) + 212 = 299 + 212 = 511\,\text{km}$. The residual is $e = y - \hat{y} = 501 - 511 = -10\,\text{km}$.

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19 [Challenging] • [FRQ] • Target: Point-to-Line Distance & Tangent Circle • Cross-Lesson: Topic 9.01 (Circles & Coordinate Geometry)

The line $y = \frac{3}{4}x + 5$ is tangent to a circle centered at $C(4, -2)$. Find the perpendicular distance from $C(4, -2)$ to the line, and write the standard equation of the circle.

Write the line in general form $3x - 4y + 20 = 0$. The perpendicular distance from $C(4, -2)$ to the tangent line equals the radius r: $r = \frac{|3(4) - 4(-2) + 20|}{\sqrt{3^2 + (-4)^2}} = \frac{|12 + 8 + 20|}{5} = \frac{40}{5} = 8$. Thus the circle has equation $(x - 4)^2 + (y + 2)^2 = 64$.

20 [Challenging] • [FRQ] • Target: Linear Calibration & Percentage Rate Comparison • Cross-Lesson: Topic 5.01 (Ratios, Rates & Percentages)

In a laboratory calibration, temperature C (in degrees Celsius) and resistance R (in ohms) are linearly related. At $C = 20$, $R = 108.0\,\Omega$; at $C = 95$, $R = 138.0\,\Omega$. Meanwhile, a second sensor has resistance $R_2(C) = 96.0 + 0.65C$. At what temperature C do both sensors have the exact same resistance, and what is the percentage increase in the first sensor's resistance from $C = 0$ to $C = 100$?

First sensor slope: $m_1 = \frac{138.0 - 108.0}{95 - 20} = \frac{30}{75} = 0.40\,\Omega/^\circ\text{C}$. Intercept: $R_1(0) = 108.0 - 0.40(20) = 100.0\,\Omega$, so $R_1(C) = 100.0 + 0.40C$. Setting $R_1(C) = R_2(C)$: $100.0 + 0.40C = 96.0 + 0.65C \Rightarrow 0.25C = 4.0 \Rightarrow C = 16^\circ\text{C}$. From $C = 0$ ($R_1 = 100\,\Omega$) to $C = 100$ ($R_1 = 140\,\Omega$), the percentage increase is $\frac{140 - 100}{100} \times 100\% = 40\%$.

Part II — Multiple-Choice Questions (MCQs): Foundational Concepts (21–28)

21 [Foundational] • [MCQ] • Target: Structure in Linear Equations

If $4(2x - 3) + 5 = 3x + 18$, what is the value of $5x - 7$?

a. 18 ✓ **b.** 23

c. 25 **d.** 30

Expand: $8x - 12 + 5 = 3x + 18 \Rightarrow 5x - 7 = 18$. Notice the question asks directly for $5x - 7$, which equals 18 without even needing to divide by 5 ($x = 5$).

22 [Foundational] • [MCQ] • Target: Point-Slope & Standard Form

Which of the following is an equation of the line in the xy-plane that has slope $-\frac{3}{5}$ and passes through $(10, -2)$?

a. $3x + 5y = 20$ ✓ **b.** $3x - 5y = 40$

c. $5x + 3y = 44$ **d.** $3x + 5y = -20$

Using point-slope form: $y - (-2) = -\frac{3}{5}(x - 10) \Rightarrow 5(y + 2) = -3x + 30 \Rightarrow 3x + 5y = 20$.

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23 [Foundational] • [MCQ] • Target: Intercepts from Intercept Form

What is the x-intercept of the graph of $\frac{x}{6} - \frac{y}{4} = 3$ in the xy-plane?

a. $(-12, 0)$

b. $(6, 0)$

c. $(12, 0)$

d. $(18, 0)$ ✓

Set $y = 0$: $\frac{x}{6} = 3 \Rightarrow x = 18$, so the x-intercept is $(18, 0)$.

24 [Foundational] • [MCQ] • Target: Interpreting Slope in Context

A botanist models the height $h(t)$, in centimeters, of a bamboo shoot t days after planting by $h(t) = 4.5t + 12$. Which statement best interprets the number 4.5 in this context?

a. The initial height of the shoot was 4.5 cm.

b. **The shoot grows an average of 4.5 cm per day.** ✓

c. The shoot reaches 12 cm after 4.5 days.

d. The height increases by 12 cm every 4.5 days.

In the linear model $h(t) = mt + b$, the slope $m = 4.5$ represents the constant rate of change of height with respect to time: 4.5 cm per day.

25 [Foundational] • [MCQ] • Target: Perpendicular Slopes

Line p is defined by $2y - 6x = 9$. Line q is perpendicular to line p in the xy-plane. What is the slope of line q?

a. -3

b. $-\frac{1}{3}$ ✓

c. $\frac{1}{3}$

d. 3

Rewrite line p as $y = 3x + \frac{9}{2}$, which has slope $m_p = 3$. A perpendicular line has negative reciprocal slope $m_q = -\frac{1}{3}$.

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26 [Foundational] • [MCQ] • Target: Rearranging Multivariable Formulas

The kinetic energy relation with linear drag work is $E = E_0 - fd$, where E_0 is initial energy, f is friction force, and d is distance. Which equation correctly expresses d in terms of $E, E_0,$ and f?

a. $d = \frac{E - E_0}{f}$

b. $d = \frac{E_0 - E}{f}$ ✓

c. $d = f(E_0 - E)$

d. $d = E_0 - \frac{E}{f}$

Add fd and subtract E: $fd = E_0 - E \Rightarrow d = \frac{E_0 - E}{f}$.

27 [Foundational] • [MCQ] • Target: Linear Equations with Fractional Coefficients

Solve the linear fractional equation $\frac{2x+5}{3} - \frac{x-1}{2} = 4$ for the unknown x.

a. 7

b. 11 ✓

c. 13

d. 17

Multiply both sides by 6: $2(2x+5) - 3(x-1) = 24 \Rightarrow 4x + 10 - 3x + 3 = 24 \Rightarrow x + 13 = 24 \Rightarrow x = 11$.

28 [Foundational] • [MCQ] • Target: Reading & Extrapolating Linear Graphs

The graph of a linear function f is shown below. What is the value of $f(8)$?

a. -7

b. -5 ✓

c. -3

d. -1

From the plotted graph, the line passes through $(0, 3)$ and $(4, -1)$, so its slope is $m = \frac{-1-3}{4-0} = -1$ and $f(x) = -x + 3$. Therefore $f(8) = -8 + 3 = -5$.

Part II — Multiple-Choice Questions (MCQs): Intermediate Exam Practice (29–37)

29 [Intermediate] • [MCQ] • Target: No-Solution Parameter Conditions

In the equation $k(3x - 4) = 6(2x - 5) + 2$, k is a constant. If the equation has no solution, what is the value of k?

a. 2

b. 3

c. 4 ✓

d. 6

Expand: $3kx - 4k = 12x - 28$. For no solution, the x-coefficients must be equal ($3k = 12 \Rightarrow k = 4$) while the constant terms differ ($-4(4) = -16 \neq -28$). Thus $k = 4$.

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30 [Intermediate] • [MCQ] • Target: Piecewise Linear Word Problems

A catering company charges 450 dollars for up to 20 guests, plus 18.50 dollars for each guest beyond 20. If a client's total bill before tax was 968 dollars, how many guests attended the event?

a. 28

b. 44

c. 48 ✓

d. 52

Let $g > 20$ be the total number of guests: $450 + 18.50(g - 20) = 968 \Rightarrow 18.50(g - 20) = 518 \Rightarrow g - 20 = 28 \Rightarrow g = 48$ guests.

31 [Intermediate] • [MCQ] • Target: Finding Intercepts from Function Values

If f is a linear function such that $f(-2) = 11$ and $f(6) = -5$, what is the x-intercept of the graph of $y = f(x)$?

a. $(2.5, 0)$

b. $(3, 0)$

c. $(3.5, 0)$ ✓

d. $(4, 0)$

Slope $m = \frac{-5 - 11}{6 - (-2)} = \frac{-16}{8} = -2$. Then $f(x) = -2(x - 6) - 5 = -2x + 7$. Setting $f(x) = 0$ yields $x = 3.5$.

32 [Intermediate] • [MCQ] • Target: Absolute Value Non-Negativity

How many real solutions does the equation $|4x - 9| + 6 = 2$ have?

a. **Zero** ✓

b. One

c. Two

d. Infinitely many

Subtract 6 from both sides: $|4x - 9| = -4$. Because the absolute value of any real expression is non-negative, there are zero real solutions.

33 [Intermediate] • [MCQ] • Target: Sum of Solutions of Absolute Value Equations

What is the sum of all solutions to the equation $2|3x - 6| = 18$?

a. 2

b. 4 ✓

c. 5

d. 6

Divide by 2: $|3x - 6| = 9 \Rightarrow 3x - 6 = 9$ ($x = 5$) or $3x - 6 = -9$ ($x = -1$). The sum of the solutions is $5 + (-1) = 4$.

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34 [Intermediate] • [MCQ] • Target: Algebraic Slope Coordinates

Line L passes through $(k, 3)$ and $(9, 2k-1)$ in the xy-plane and has slope $\frac{3}{2}$. What is the value of k?

a. 4

b. 5 ✓

c. 6

d. 7

Use the slope formula: $\frac{(2k-1)-3}{9-k} = \frac{3}{2} \Rightarrow 2(2k-4) = 3(9-k) \Rightarrow 4k-8 = 27-3k \Rightarrow 7k = 35 \Rightarrow k = 5$.

35 [Intermediate] • [MCQ] • Target: Intercept Form & Parametric Slope

The line $\frac{x}{a} + \frac{y}{b} = 1$ passes through $(3, 4)$ and has slope $-\frac{2}{3}$. What is the value of $a + b$?

a. 12

b. 15 ✓

c. 16

d. 18

The slope of $\frac{x}{a} + \frac{y}{b} = 1$ is $-\frac{b}{a} = -\frac{2}{3}$, so $b = \frac{2}{3}a$. Substituting $(3, 4)$: $\frac{3}{a} + \frac{4}{(2/3)a} = 1 \Rightarrow \frac{3}{a} + \frac{6}{a} = 1 \Rightarrow a = 9$, and $b = \frac{2}{3}(9) = 6$. Thus $a + b = 9 + 6 = 15$.

36 [Intermediate] • [MCQ] • Target: Translations of Linear Graphs

When the graph of $y = 4x - 7$ is translated 3 units to the left and 5 units upward in the xy-plane, what is the y-intercept of the resulting line?

a. $(0, 5)$

b. $(0, 7)$

c. $(0, 10)$ ✓

d. $(0, 12)$

Translating 3 units left replaces x with $x + 3$, and 5 units upward adds 5:
$y = 4(x + 3) - 7 + 5 = 4x + 12 - 2 = 4x + 10$. The new y-intercept is $(0, 10)$.

37 [Intermediate] • [MCQ] • Target: Multi-Step Linear Discount & Tax Models

A store discounts a jacket by 20% off its original price p, and a customer then applies a 15-dollar coupon before paying 8% sales tax on the reduced amount. If the final total is 108.00 dollars, what was the original price p?

a. 135.00 dollars

b. 140.00 dollars

c. 143.75 **dollars** ✓

d. 150.00 dollars

Set up the linear equation: $1.08(0.80p - 15) = 108.00 \Rightarrow 0.80p - 15 = 100.00 \Rightarrow 0.80p = 115.00 \Rightarrow p = 143.75$ dollars.

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Part II — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging] • [MCQ] • Target: Area Bounded by Intersecting Lines • Cross-Lesson: Topic 2.01 (Linear Systems) & Topic 7.01 (Triangle Area)**

What is the area of the triangle enclosed by the lines $y = 2x + 4$, $y = -x + 7$, and the x-axis?

a. 13.5

b. 18

c. 24

d. 27 ✓

Find the vertices: $y = 2x + 4$ meets the x-axis at $(-2, 0)$; $y = -x + 7$ meets the x-axis at $(7, 0)$. The two lines intersect where $2x + 4 = -x + 7 \Rightarrow 3x = 3 \Rightarrow x = 1, y = 6$. The base along the x-axis is $7 - (-2) = 9$ and the altitude is $y = 6$, so Area $= \frac{1}{2}(9)(6) = 27$.

39 **[Challenging] • [MCQ] • Target: Perpendicular Bisector in Coordinate Geometry • Cross-Lesson: Topic 7.01 (Geometry & Midpoint Theorem)**

In the xy-plane, the perpendicular bisector of the segment joining $A(-2, 1)$ and $B(6, 5)$ intersects the y-axis at $(0, b)$. What is the value of b?

a. 5

b. 6

c. 7 ✓

d. 8

The midpoint of AB is $M = \left(\frac{-2+6}{2}, \frac{1+5}{2}\right) = (2, 3)$. The slope of AB is $m_{AB} = \frac{5-1}{6-(-2)} = \frac{1}{2}$, so the perpendicular bisector has slope $m_{\perp} = -2$ and equation $y - 3 = -2(x - 2) \Rightarrow y = -2x + 7$. Thus $b = 7$.

40 **[Challenging] • [MCQ] • Target: Linear Functional Equations • Cross-Lesson: Topic 2.01 (Systems of Linear Equations)**

For all real numbers x, the linear function g satisfies $2g(x) + g(4 - x) = 6x - 3$. What is the value of $g(5)$?

a. 15

b. 18

c. 21 ✓

d. 24

Substitute $x = 5$: $2g(5) + g(-1) = 27$. Substitute $x = -1$: $2g(-1) + g(5) = -9$. Multiply the first equation by 2 and subtract the second: $3g(5) = 54 - (-9) = 63 \Rightarrow g(5) = 21$.

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41 [Challenging] • [MCQ] • Target: Tangent Line to a Parabola • Cross-Lesson: Topic 3.01 (Quadratics & Discriminant Analysis)

The line $y = mx + 5$ intersects the parabola $y = x^2 + 2x + 9$ at exactly one point in the xy-plane. If $m > 0$, what is the value of m?

a. 4

b. 6 ✓

c. 8

d. 10

Set $x^2 + 2x + 9 = mx + 5 \Rightarrow x^2 + (2 - m)x + 4 = 0$. For tangency (one intersection), the discriminant must be zero: $\Delta = (2 - m)^2 - 4(1)(4) = 0 \Rightarrow 2 - m = \pm 4$. Since $m > 0$, $2 - m = -4 \Rightarrow m = 6$.

42 [Challenging] • [MCQ] • Target: Absolute Value Interval Reconstruction • Cross-Lesson: Topic 2.01 (Linear & Absolute Value Inequalities)

If the solution set to the inequality $|3x - a| \leq b$ is the closed interval $[-2, 6]$, what is the value of $a + b$?

a. 12

b. 14

c. 18 ✓

d. 20

The interval $[-2, 6]$ has midpoint $c = \frac{-2+6}{2} = 2$ and half-width $r = \frac{6-(-2)}{2} = 4$. Rewriting $|3x - a| \leq b$ as $|x - \frac{a}{3}| \leq \frac{b}{3}$ gives $\frac{a}{3} = 2 \Rightarrow a = 6$ and $\frac{b}{3} = 4 \Rightarrow b = 12$. Thus $a + b = 18$.

43 [Challenging] • [MCQ] • Target: Linear Drainage Rate in a 3D Cylinder • Cross-Lesson: Topic 8.01 (Area, Volume & 3D Solids)

A cylindrical water tank of radius $r = 2\,\text{m}$ and height $H = 10\,\text{m}$ is initially full and drains at a constant volumetric rate so that the water depth $h(t)$ decreases linearly by $0.4\,\text{m}$ per minute. After how many minutes does the tank contain $24\pi\,\text{m}^3$ of water?

a. 8

b. 10 ✓

c. 12

d. 15

The volume of water at depth h is $V = \pi r^2 h = 4\pi h$. When $V = 24\pi$, the depth is $h = \frac{24\pi}{4\pi} = 6\,\text{m}$. Since $h(t) = 10 - 0.4t$, solving $10 - 0.4t = 6$ yields $0.4t = 4 \Rightarrow t = 10$ minutes.

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44 [Challenging] • [MCQ] • Target: Coordinate Reflection Across an Oblique Line • Cross-Lesson: Topic 7.01 (Coordinate Transformations)

The reflection of the point $P(1, 7)$ across the line $y = x + 2$ has coordinates (x_0, y_0). What is the value of $x_0 + y_0$?

a. 6

b. 8 ✓

c. 10

d. 12

The reflection of (x_1, y_1) across $y = x + 2$ (or $x - y = -2$) is $(y_1 - 2, x_1 + 2) = (7 - 2, 1 + 2) = (5, 3)$. Check: midpoint is $(3, 5)$, which satisfies $5 = 3 + 2$, and slope is $\frac{3-7}{5-1} = -1 \perp 1$. Thus $x_0 + y_0 = 5 + 3 = 8$.

45 [Challenging] • [MCQ] • Target: Effect of Linear Transformations on Spread • Cross-Lesson: Topic 6.01 (Statistics: Mean & Standard Deviation)

In a data set of 20 observations (x_i, y_i), every point lies on the line $y = -2.5x + 14$. If the standard deviation of the x-values is $s_x = 3.2$, what is the standard deviation s_y of the y-values?

a. -8.0

b. 5.6

c. 8.0 ✓

d. 22.0

Under a linear transformation $y = mx + b$, the standard deviation scales by $|m|$ and is unaffected by the vertical shift b: $s_y = |-2.5|s_x = 2.5(3.2) = 8.0$.

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